

Solutions – Quiz 5

ECON 320 – Introduction to Mathematical Economics

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October 29, 2025

Multiple Choice (1 mark each)

Q1. Correct: (b)

Reasoning: The Hessian of $f(x, y)$ is the 2×2 matrix of second partial derivatives

$$H(f) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

Option (a) describes the Jacobian of first derivatives; (c) “total differentials” is not a matrix of second partials; (d) the Jacobian applies to vector-valued functions.

Q2. Correct: (b)

Reasoning: For two variables, define $D = f_{xx}f_{yy} - (f_{xy})^2$. If at a critical point we have $D > 0$ and $f_{xx} < 0$, then the Hessian is negative definite $\Rightarrow$ a **local maximum**. (If $D > 0$ and $f_{xx} > 0$, it is a local minimum. If $D < 0$, it is a saddle.)

Q3. Correct: (d)

Problem: $f(x, y) = x^2 - 2xy + 3y^2$. Then $f_{xx} = 2$, $f_{yy} = 6$, $f_{xy} = -2$. Hence

$$D = f_{xx}f_{yy} - (f_{xy})^2 = 2 \cdot 6 - (-2)^2 = 12 - 4 = 8 > 0.$$

So the determinant is 8.

Q4. Correct: (c)

Reasoning: For a **local maximum**, the Hessian must be negative definite: $D > 0$ and $f_{xx} < 0$. (For a local minimum: $D > 0$ and $f_{xx} > 0$; $D < 0$ indicates a saddle.)

Q5. Correct: (d)

Problem: $f(x, y) = 4x^2 + 9y^2 - 12xy$. Then $f_{xx} = 8$, $f_{yy} = 18$, $f_{xy} = -12$. Thus

$$D = 8 \cdot 18 - (-12)^2 = 144 - 144 = 0.$$

Hence the second-order test is *inconclusive*. In fact, $f(x, y) = (2x - 3y)^2 \geq 0$, so $(0, 0)$ is a (non-strict) minimum and the set of minimizers is the line $2x = 3y$. Among the given choices, only (d) is unambiguously true.

Problem 1 (6 marks)**Function:** $f(x, y) = 3x^2 + xy + y^2 - 6x - 9y$.(a) Stationary points. First-order conditions:

$$f_x = 6x + y - 6 = 0, \quad f_y = x + 2y - 9 = 0.$$

Solve the linear system:

$$\begin{cases} 6x + y = 6 \\ x + 2y = 9 \end{cases} \Rightarrow \text{From the first, } y = 6 - 6x.$$

Substitute into the second:

$$x + 2(6 - 6x) = 9 \Rightarrow x + 12 - 12x = 9 \Rightarrow -11x = -3 \Rightarrow x^* = \frac{3}{11}.$$

Then

$$y^* = 6 - 6x^* = 6 - \frac{18}{11} = \frac{66 - 18}{11} = \frac{48}{11}.$$

Thus the unique stationary point is

$$(x^*, y^*) = \left(\frac{3}{11}, \frac{48}{11} \right).$$

(b) Nature of the point (Hessian test). Compute second partials:

$$f_{xx} = 6, \quad f_{yy} = 2, \quad f_{xy} = 1.$$

Then the Hessian determinant

$$D = f_{xx}f_{yy} - (f_{xy})^2 = 6 \cdot 2 - 1^2 = 12 - 1 = 11 > 0,$$

and $f_{xx} = 6 > 0$. Therefore, the Hessian is *positive definite* at the stationary point:

$$\text{Local minimum at } \left(\frac{3}{11}, \frac{48}{11} \right).$$

(Optional check: the minimized value is $f\left(\frac{3}{11}, \frac{48}{11}\right) = 3\left(\frac{9}{121}\right) + \left(\frac{144}{121}\right) + \left(\frac{2304}{121}\right) - 6\left(\frac{3}{11}\right) - 9\left(\frac{48}{11}\right)$, which simplifies to a positive number, consistent with a minimum.)

Marking guide (6): 2 for FOCs, 2 for solving the system, 2 for correct Hessian test and classification.

Problem 2 (9 marks)**Cost minimization with target output.**

Production $Q = 100K^{0.5}L^{0.5}$, input prices $r = 4$ (capital), $w = 1$ (labor), target $Q_0 = 100$.

Set-up: Minimize $C = 4K + 1 \cdot L$ subject to $100K^{0.5}L^{0.5} = 100$.

Constraint simplification:

$$100K^{0.5}L^{0.5} = 100 \iff K^{0.5}L^{0.5} = 1 \iff KL = 1.$$

Lagrangian:

$$\mathcal{L}(K, L, \lambda) = 4K + L + \lambda(100 - 100K^{0.5}L^{0.5}).$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial K} = 4 - 50\lambda L^{0.5}K^{-0.5} = 0 \implies 4 = 50\lambda \frac{\sqrt{L}}{\sqrt{K}} \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial L} = 1 - 50\lambda K^{0.5}L^{-0.5} = 0 \implies 1 = 50\lambda \frac{\sqrt{K}}{\sqrt{L}} \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} : 100 - 100K^{0.5}L^{0.5} = 0 \iff KL = 1. \quad (3)$$

Ratio condition (MRTS = price ratio): Divide (1) by (2):

$$\frac{4}{1} = \frac{50\lambda(\sqrt{L}/\sqrt{K})}{50\lambda(\sqrt{K}/\sqrt{L})} = \frac{L}{K} \implies L = 4K.$$

This is equivalent to the standard Cobb–Douglas rule $\frac{\text{MPL}}{\text{MPK}} = \frac{w}{r} = \frac{1}{4}$ and indeed $\frac{\text{MPL}}{\text{MPK}} = \frac{K}{L}$, giving $K/L = 1/4$ (i.e., $L = 4K$).

Use the constraint $KL = 1$:

$$K \cdot (4K) = 1 \implies 4K^2 = 1 \implies K^* = \frac{1}{2}, \quad L^* = 2.$$

Minimum cost:

$$C_{\min} = 4K^* + 1 \cdot L^* = 4\left(\frac{1}{2}\right) + 2 = 2 + 2 = \boxed{4}.$$

Second-order condition (brief): With a convex cost function and a concave (Cobb–Douglas) production constraint in (K, L) at fixed Q , the problem is convex; the FOCs are sufficient. Alternatively, the bordered Hessian test confirms a minimum.

$K^* = \frac{1}{2}, \quad L^* = 2, \quad C_{\min} = 4.$

Marking guide (8): 2 for Lagrangian & FOCs, 2 for ratio condition, 2 for applying the constraint, 1 for optimal bundle, 1 for cost.

Total Section B: 15/15 (Overall quiz 20/20).